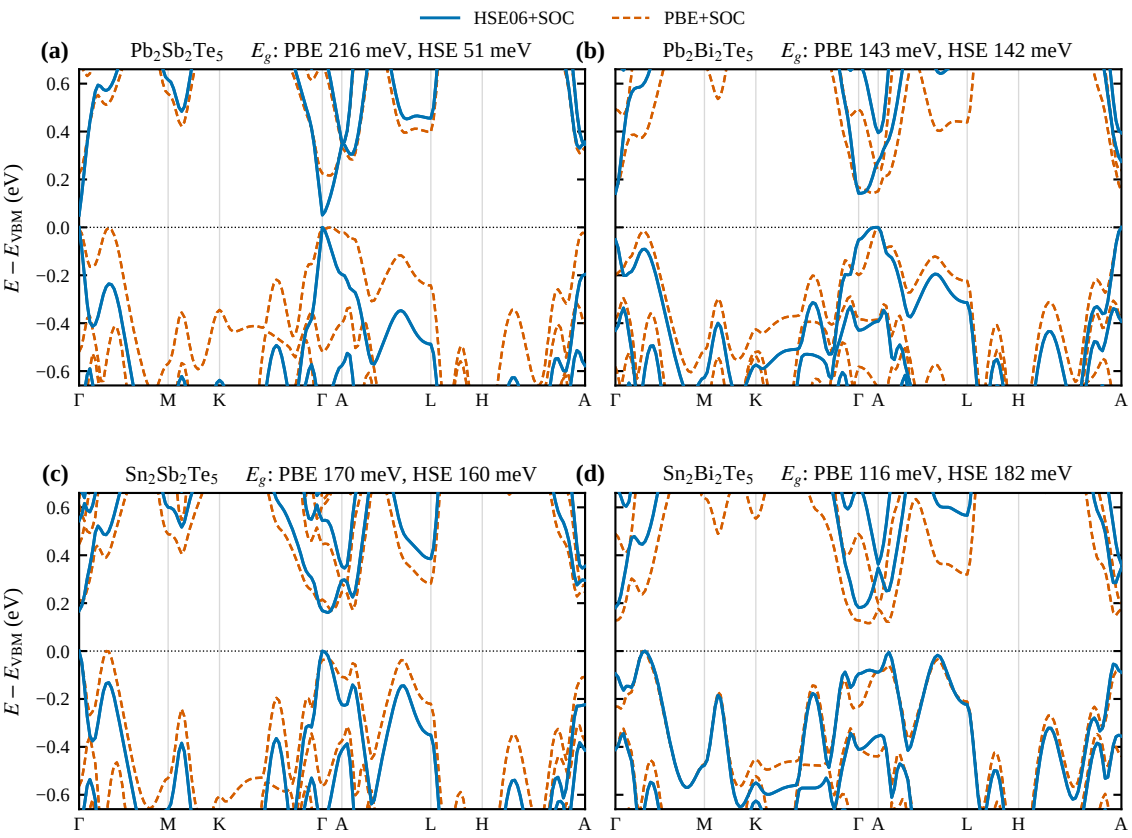

HSE06+SOC
PBE+SOC
(a)
$Pb_2Sb_2Te_5$ $E_g$: PBE 216 meV, HSE 51 meV
(b)
$Pb_2Bi_2Te_5$ $E_g$: PBE 143 meV, HSE 142 meV
(c)
$Sn_2Sb_2Te_5$ $E_g$: PBE 170 meV, HSE 160 meV
(d)
$Sn_2Bi_2Te_5$ $E_g$: PBE 116 meV, HSE 182 meV
$E - E_{VBM}$ (eV)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Γ
M
K
Γ
A
L
H
A